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ARTICLE

Population pharmacokinetic modeling of sotatercept in healthy participants and patients with pulmonary arterial hypertension

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Abstract

Sotatercept is a breakthrough, first-in-class biologic, that is FDA-approved for the treatment of pulmonary arterial hypertension (PAH). A population pharmacokinetic (PopPK) model was developed using data from two phase 1 studies in healthy participants, and two phase 2 studies and one phase 3 study in participants with PAH. The pooled sotatercept PK data encompassed single intravenous (IV) or subcutaneous (SC) doses ranging from 0.01 to 3.0 mg/kg, as well as multiple SC doses ranging from 0.03 to 1.0 mg/kg, with PK samples collected up to a maximum of ~150 weeks following Q3W and Q4W dosing regimens. The final PopPK analysis included 350 participants, with 30 and 320 participants receiving sotatercept IV and SC, respectively. A two-compartment model with a first-order absorption rate constant and a linear disposition from central compartment well-described sotatercept PK. The estimated bioavailability is ~66%; bioavailability, clearance (CL), and central volume (VC) have low to moderate inter-individual variability. Time-varying body weight and baseline albumin concentration were statistically significant predictors of PK; CL and VC were predicted to increase with increasing body weight, while CL was predicted to decrease with increasing baseline albumin concentration. However, the magnitude of covariate effects is not predicted to meaningfully alter the disposition of sotatercept. Altogether, the PopPK modeling results demonstrate favorable PK characteristics (low to moderate variability and typical bioavailability), supporting sotatercept as a SC biological agent for the treatment of patients with PAH.

Study Highlights

WHAT IS THE CURRENT KNOWLEDGE ON THE TOPIC?

PAH is a rare disease causing a progressive increase in pulmonary vascular resistance resulting in right ventricular dysfunction, heart failure, and

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ultimately premature death. Currently, available therapies are solely symptomatic. Sotatercept is the first breakthrough, first-in-class biologic, FDA-approved for the treatment of PAH.

WHAT QUESTION DID THIS STUDY ADDRESS?

Using nonlinear mixed-effect modeling approach, a two-compartment PopPK model with first-order absorption and linear disposition characterized sotatercept serum concentration data in healthy participants from 2 phase 1 studies and patients with PAH from 2 phase 2 studies and 1 phase 3 study. In all studies, sotatercept was administered IV or SC, as single or multiple doses (Q3W and Q4W) ranging from 0.01 to 3.0 mg/kg. Participants were sampled for a maximum of ~150 weeks.

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

Sotatercept inter-individual PK variability is moderate. Time-varying BW and baseline albumin concentrations were statistically significant predictors of sotatercept PK; with CL and VC predicted to increase with increasing BW, and CL predicted to decrease with increasing baseline albumin concentration. Sotatercept bioavailability estimated at ~66% is typical of that of a biologic.

HOW MIGHT THIS CHANGE DRUG DISCOVERY, DEVELOPMENT, AND/OR THERAPEUTICS?

This study has improved the scientific understanding of sotatercept PK and showed favorable PK characteristics (low to moderate variability and typical bioavailability) supporting sotatercept as a SC biological agent for the treatment of patients with PAH.

INTRODUCTION

Pulmonary arterial hypertension (PAH) is a rare, progressive, and life-threatening disorder of the lungs.[1] Its prevalence is estimated at 10–52 cases per million and the mean age at diagnosis is between 43 and 46 years with a higher proportion of female patients.[2–8] PAH is characterized by a narrowing of pulmonary arteries that increases the blood pressure within the lungs and restricts the flow of blood from the heart to the lungs. Symptoms of PAH include shortness of breath, fatigue, chest pain, and swelling in the legs and ankles. As the disease progresses, the increase in blood pressure can lead to right ventricle heart failure and death. At the molecular level, an imbalance in signaling pathways that promotes hyperproliferation of cells in the endothelium is a major factor responsible for the pathology underlying PAH.[9,10]

The goal of a treatment in PAH is to improve pulmonary hemodynamics, exercise capacity, and progression-free survival. The current approved treatments work primarily through vasodilation. They include phosphodiesterase-5 inhibitors, endothelin-receptor antagonists, soluble guanylate cyclase stimulators, and compounds targeting the prostacyclin pathway.[11–13] However, many of these treatments do not achieve the PAH treatment goal. A key unmet therapeutic need in PAH is to develop disease-modifying agents that target the underlying root cause of PAH by restoring the homeostasis of the vascular-wall in the lungs.

Sotatercept is a novel breakthrough and first-in-class biologic therapeutic that is FDA-approved for the treatment of PAH. It is a recombinant fusion protein consisting of the extracellular domain of the activin receptor type IIA (ActRIIA) linked to the human IgG1 Fc domain. Sotatercept acts as a ligand trap for activin A and other ligands for ActRIIA, consequently, inhibiting the activin signaling pathway. In addition, sotatercept likely reduces the production of pro-inflammatory molecules and increases the production of anti-inflammatory molecules, which may reduce the inflammation associated with PAH. Further, sotatercept has been shown to reduce the levels of endothelin-1, a molecule that is involved in the development of PAH. Sotatercept may also improve pulmonary vascular remodeling by restoring balance to the Smad signaling pathway.[14]

In clinical trials, sotatercept reduced pulmonary vascular resistance (PVR) and improved exercise capacity, measured by the 6-minute walk distance (6MWD), in participants with PAH. In a phase 2 trial, sotatercept was found to reduce PVR by an average of 20.5% in the 0.3 mg/kg Q3W arm and 33.9% in the 0.7 mg/kg Q3W arm after 6 months of treatment, while for 6 MWD both 0.3 and

0.7 mg/kg arms were similar with 14.3% improvement from baseline at 6 months (10% improvement relative to placebo).[15] In a phase 3 trial, sotatercept was found to reduce PVR by an average of 25% and improve exercise capacity by an average of 15% after 24 weeks of treatment. Additionally, sotatercept was found to be well-tolerated with no serious adverse events reported.[16] We developed a population pharmacokinetic (PopPK) model leveraging sotatercept PK data from two phase 1 trials including a single ascending dose study (SAD) and a multiple ascending dose study (MAD); two phase 2 studies, and one phase 3 trial. The objectives of this integrated modeling were to characterize the PK of sotatercept in PAH patients and to identify covariates that may influence sotatercept systemic exposure.

METHODS

Clinical trial designs and populations

Data used to conduct the PopPK modeling included two phase 1 trials in healthy post-menopausal women (PMW), two phase 2 trials (SPECTRA and PULSAR), and one phase 3 trial (STELLAR), conducted in participants with PAH. These trials were conducted according to the principles of Good Clinical Practice and the study protocols were reviewed and approved by the Institutional Review Board or Independent Ethics Committee at each study site. All participants provided informed consent prior to the conduct of the trials.

The two phase 1 trials were a SAD study and a MAD study. Rich PK data were collected in SAD and MAD studies and sparse PK data were collected in SPECTRA, PULSAR, and STELLAR (Figure 1). Furthermore, details on the dosing regimens and sampling schemes for all studies included in this analysis are provided in Supplementary Material S1.

Data analyses and population pharmacokinetic modeling approach

Data analyses were performed using SAS Version 9.4 and KIWI Version 2022R1. Population modeling was performed using the computer program NONMEM®, Version 7, Level 3.0. NONMEM analyses were performed on an Intel cluster with the Linux operating system. The general steps that were followed for the development of the sotatercept population PK (PopPK) model was: (1) exploratory data analysis, (2) initial base structural model development and evaluation of covariate effects (SAD trial, MAD trial, and preliminary data from PULSAR and SPECTRA), (3) model update and refinement (SAD trial, MAD trial; final data from PULSAR, SPECTRA, and STELLAR), and (4) model qualification.

Exploratory data analyses showed that sotatercept PK followed a two-compartment model with linear elimination (CL), first-order absorption rate constant (KA) for the subcutaneous (SC) administration, and data-defined rates for intravenous (IV) infusions. A PopPK model was developed to fit the data by the estimation of typical structural model parameters including KA, central volume (VC), peripheral volume (VP), CL, distribution clearance (Q), magnitude of inter-individual variability (IIV), and residual variability (RV). The model for IIV was described with exponential error models for all PK parameters except the bioavailability (F1). The exponential error model is expressed as the following: $X_i = \widetilde{X}_i \times \exp\left(\eta_i^x\right)$, where Xi is the individual-specific estimate of the X parameter in the ith participant; $\widetilde{X}_i$is the typical value of the X parameter in the ith participant; and η_i^x is a random variable that represents the persistent difference between the "true" individual-specific estimate and the typical value of the X parameter in the ith participant; η_i^x are assumed to be independent, identically distributed statistical errors with a mean of 0 and a variance of $\omega_x^2\omega_x^2$. The estimates are presented as coefficients of variation expressed as a percent (%CV), which were approximated as $\%\,\mathrm{CVx} = \sqrt{\exp\left(\omega_x^2\right) - 1} \times 100$.

For F1, a logit error model was employed to restrain the range within [0,1] $X_i = \frac{e^{\log\left(\frac{\%\tilde{X}_i}{1-\%\tilde{X}_i}\right)+\eta_i^x}}{1+e^{\log\left(\frac{\%\tilde{X}_i}{1-\%\tilde{X}_i}\right)+\eta_i^x}}$, where X_i is the individual-specific estimate of the X parameter in the ith participant, $\widetilde{X}_i$ is the typical value of the X parameter in the ith participant, η_i^x is a random variable that represents the persistent difference between the "true" individual-specific estimate and the typical value of the X parameter in the ith participant, η_i^x are assumed to be independent, identically distributed statistical errors with a mean of 0 and a variance of ω_x^2. Estimates of IIV in F1 are presented as $\%\,\mathrm{CV}_x = \left(1 - \%\tilde{X}_i\right) \times \sqrt{\omega_x^2} \times 100$. Below the lower limit of quantitation (BLQ) samples represented less than 5% of the total samples in the dataset. Due to this low frequency of BLQ values, the M1 method (exclusion of BLQ values from analyses) was considered to be appropriate.[17] The analyses were performed using NONMEM with the first-order conditional estimation method with interaction. The goodness-of-fit plots were examined for potential biases in the selected RV model and alternative models were evaluated using either constant coefficient of variation or proportional residual variability error model such as $Y_{ij} = \hat{Y}_{ij} \times \left(1 + \varepsilon_{ij}\right)$. The proportional model for RV assumes that the variance increases in proportion to the predicted values with $\%\,\mathrm{CV}_{\mathrm{RV}} = \sqrt{\sigma_x^2} \times 100$. For the Log error

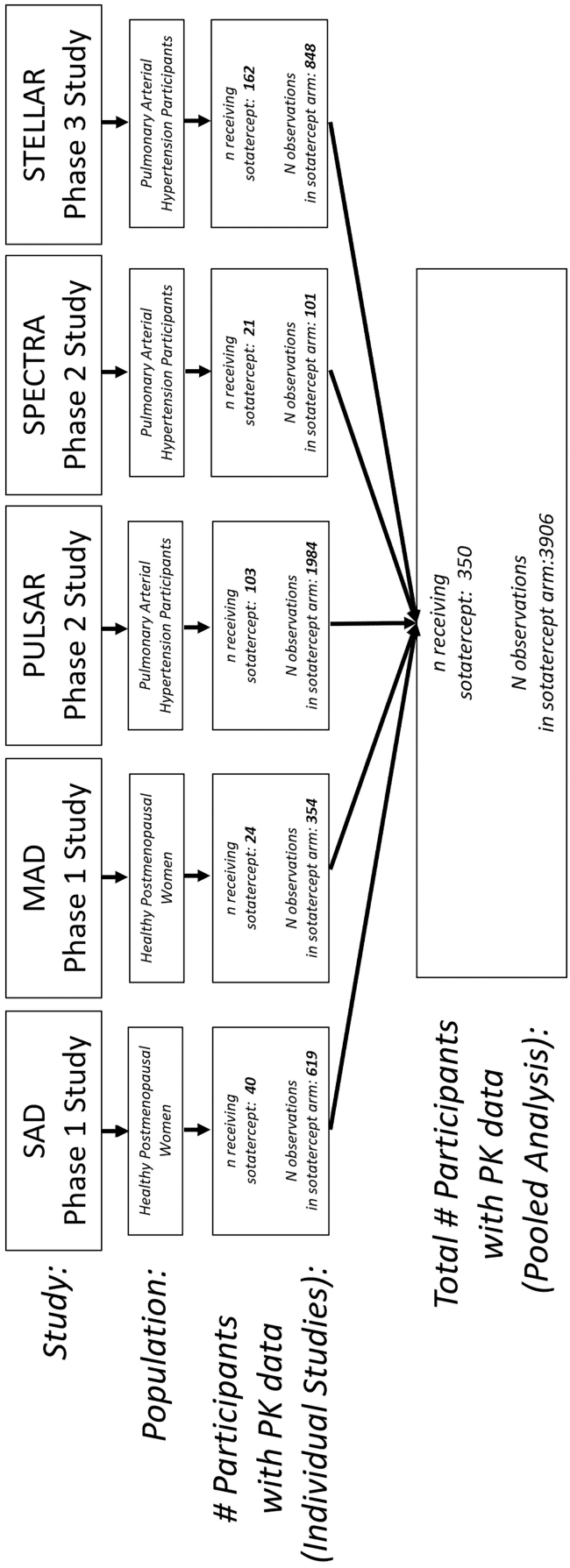

FIGURE 1 General schematic of all clinical trials included in the population PK modeling analyses. SAD is single ascending dose and MAD is multiple ascending dose.

model $\log(Y_{ij}) = \log\left(\hat{Y}_{ij}\right) + \varepsilon_{ij}$, the RV assumes a constant variance with respect to the log of the predicted value. This model is mathematically and numerically equivalent to the exponential error model for the estimation methods used. The estimate is expressed as a standard deviation (SD) (σ) in log-transformed concentration units. Where Y_{ij} is the jth measured value in the ith participant; $\hat{Y}_{ij}$ is the jth predicted value in the ith participant using the specified model; and ε_{ij} is a random variable for the discrepancy between the jth measured value in the ith participant and the predicted value; the ε_{ij} are assumed to be independent, identically distributed statistical errors with a mean of 0 and a variance of σ^2.

Assessing the impact of covariates

Continuous and categorical covariates that were tested during model development are presented in Table 1. Both power and linear relationships were tested for continuous covariates. The choice of covariates tested was based on scientific plausibility, prior knowledge, and initial data exploration.

Forward selection and backward elimination procedure

A stepwise forward selection of covariates was performed. Each covariate–parameter relationship was tested for statistical significance. Based on $\alpha = 0.01$, covariates decreasing the minimum value of the objective function (VOF) of ≥6.64 ($p < 0.01$, 1 df) and resulting in a decrease in IIV of ≥5% in the parameter of interest were considered significant.

A stepwise backward elimination proceeded after any adjustments/refinements were made to the full multivariable model following the completion of the forward selection process. A covariate was considered significant if it resulted in a decrease in the VOF ≥10.83 ($p < 0.001$, 1 df for Chi^2-distribution) when removed from the model. The backward elimination procedure was repeated until all remaining covariates were significant.

Clinical relevance of covariates

The clinical relevance of statistically significant covariate effects was assessed through the summarization and/or graphical representation of individual-specific exposure measures and parameter values vs. covariates.

To characterize the expected impact of covariates on sotatercept PK, simulations were performed based on the final integrated phase 1/2/3 PK model, individual Bayesian estimates of PK parameters, and individual values of the tested covariates assuming hypothetical SC dosing of 0.7 mg/kg every 21 days for 24 weeks in participants included in the analysis dataset. Numerical integration was performed in NONMEM to predict sotatercept concentration–time profiles for each participant. These were used to obtain maximum concentration, minimum concentration, the area under the effect curve from 0 to 504 h post-dose, and average concentration values for each participant. Predicted sotatercept exposures following the last SC dose (at Week 24) represented drug exposures at steady state. Summary statistics of simulated exposures were calculated and compared, stratified by covariate group (for discrete covariates) or quantile (for continuous covariates) in patients who achieved their target final dose during titration. In addition, forest plots of the geometric mean ratios (GMRs) relative to the reference category and 90% confidence intervals (CI) of exposure measures for covariates of interest were constructed to illustrate the relative differences in the impact of various covariates. The GMRs (CI) were initially compared with bioequivalence (BE) bounds (0.8, 1.25). If any exceeded BE bounds, the clinical significance would be further assessed based on the exposure–response results for safety and efficacy. Clinically meaningful impact on sotatercept PK was determined based on the GMR and 90% CI of each covariate, where the clinical criterion was the normal physiological values of each covariate.

The terminal elimination half-life ($t_{1/2}$) was calculated, such as $t_{1/2} = \frac{\ln(2)}{\beta}$, with $\beta = 0.5 \times \left[\left(k_{12} + k_{21} + k_{el}\right) - \sqrt{\left(k_{12} + k_{21} + k_{el}\right)^2 - \left(4 \times k_{21} \times k_{el}\right)}\right]$, where k_{12}, k_{21}, k_{el} represent the microconstants defined according to the estimated model parameters CL, Q, VC, and VP as $k_{12} = \frac{Q}{VC}$, $k_{21} = \frac{Q}{VP}$, and $k_{el} = \frac{CL}{VC}$.

Model qualification

The adequacy of the final model was evaluated using a simulation-based, predicted-corrected visual predictive check (pcVPC) method.[18] Using NONMEM, the final integrated phase 1/2/3 PK model was used to simulate 500 replicates of the PK analysis dataset population. Statistics of interest were calculated from the simulated and observed data for comparison; for example, the 5th, 50th (median), and 95th percentiles of the distributions of simulated and observed concentrations were

TABLE 1 Demographic characteristics of participants included in the final PK analysis dataset for the integrated phase 1/2/3 population pharmacokinetic model, by study.

Variable		SAD trial ($n = 40$)	MAD trial ($n = 24$)	PULSAR ($n = 103$)	SPECTRA ($n = 21$)	STELLAR ($n = 162$)	Overall ($n = 350$)
Age (years)	Mean (SD)	63.8 (7.9)	57.1 (7.1)	48.2 (14.5)	45.3 (15.1)	47.6 (14.1)	50.2 (14.4)
	Min, Max	51, 79	49, 81	19, 80	21, 70	18, 81	18, 81
Baseline albumin (g/dL)	Mean (SD)	4.47 (0.276)	4.32 (0.182)	4.24 (0.377)	4.31 (0.348)	4.57 (0.247)	4.43 (0.33)
	Min, Max	3.9, 5	4, 4.7	3.1, 5.1	3.7, 4.9	2.9, 5.8	2.9, 5.8
Baseline estimated glomerular filtration (mL/min/1.73 m^2)	Mean (SD)	78 (13.9)	90.1 (17.8)	79 (24.6)	73 (14.8)	91.2 (34.6)	84.9 (28.8)
	Min, Max	50, 115	52.2, 129	38.8, 190	45.1, 97.1	31.3, 255	31.3, 255
Baseline body weight (kg)	Mean (SD)	65 (11.7)	66.6 (9.82)	71.9 (18.1)	76.9 (16.8)	71.8 (19.1)	71 (17.6)
	Min, Max	46.8, 94.3	49.6, 85.7	43, 136	47.9, 108	39.6, 135	39.6, 136
Body mass index (kg/m^2)	Mean (SD)	NA (NA)	25.2 (3.16)	27 (5.65)	27.9 (4.57)	26.2 (5.71)	26.5 (5.48)
	Min, Max	NA, NA	19.9, 31.5	18.1, 48.3	17.6, 33.8	15.1, 50.3	15.1, 50.3
Racial classification, *n* (%)	White/Caucasian	17 (42.5)	15 (62.5)	95 (92.2)	17 (81)	146 (90.1)	290 (82.9)
	Black/African American	1 (2.5)	4 (16.7)	4 (3.88)	0 (0)	2 (1.23)	11 (3.14)
	Asian	17 (42.5)	5 (20.8)	0 (0)	2 (9.52)	1 (0.617)	25 (7.14)
	American Indian or Alaska Native	0 (0)	0 (0)	0 (0)	1 (4.76)	0 (0)	1 (0.286)
	Native Hawaiian, Pacific Islander	2 (5)	0 (0)	1 (0.971)	1 (4.76)	0 (0)	4 (1.14)
	Other	3 (7.5)	0 (0)	3 (2.91)	0 (0)	13 (8.02)	19 (5.43)
Renal function category, *n* (%)	Normal	9 (22.5)	10 (41.7)	27 (26.2)	2 (9.52)	68 (42)	116 (33.1)
	Mild impairment	27 (67.5)	13 (54.2)	54 (52.4)	15 (71.4)	74 (45.7)	183 (52.3)
	Moderate impairment	4 (10)	1 (4.17)	22 (21.4)	4 (19)	20 (12.3)	51 (14.6)
Sex, *n* (%)	Male	0 (0)	0 (0)	14 (13.6)	4 (19)	34 (21)	52 (14.9)
	Female	40 (100)	24 (100)	89 (86.4)	17 (81)	128 (79)	298 (85.1)

Note: In phase 2 and phase 3 studies, patients received background PAH therapy defined as patient-specific PAH medications that may consist of monotherapy or combination therapy with SOC that is endothelin-receptor antagonists, phosphodiesterase 5 inhibitors, soluble guanylate cyclase stimulators, and/or prostacyclin analogs or receptor agonists.

Abbreviations: Max, maximum; Min, minimum; n, number of participants; NA, not applicable; PK, pharmacokinetic; SD, standard deviation.

compared. These percentiles were plotted vs. time, with the original observed dataset and/or percentiles based on the observed data overlaid to visually assess concordance between the model-based simulated data and the observed data.

In addition, using the analysis dataset and final model, a bootstrap procedure was performed to assess the accuracy of the final parameter estimates. A total of 1000 bootstrap datasets were generated by resampling with replacement from the analysis dataset. Appropriate stratification variables were used. The final integrated phase 1/2/3 model was then estimated using each of the bootstrap datasets, resulting in bootstrap distributions for each of the estimated parameters. Runs that did not initially minimize were re-tried once using alternative initial estimates. The percent of datasets that successfully minimized was reported, and from these datasets, nonparametric 95% bootstrap confidence intervals were calculated for each parameter and compared with the final parameter estimates.

RESULTS

Participant characteristics

The participants included in the final PopPK model consisted of 298 females (85.1%) and 52 males (14.9%) with a mean age of 50 years ranging from 18 to 81 years. The mean body weight is 71 kg (ranging from 39.6 to 136 kg) at the baseline visit. The majority (82.9%) of participants identified as white, while 25 (7.14%) participants identified as Asian, 11 (3.14%) participants were identified as black or African American, 4 (1.14%) participants were identified as Native Hawaiian or other Pacific Islander, and 19 (5.43%) participants were identified as Other (Table 1).

Population PK model

The final PopPK model (Figure 2) was well-described by a 2-compartment model with first-order absorption and linear elimination and included exponential IIV on KA, F1, CL, VC, and VP. Two separate log error models were used to describe the RV for healthy and PAH participants.

Table 2 lists the parameter estimates of the final PopPK model along with the mean and 95% confidence intervals (CIs) from a 1000-replicate bootstrap analysis. All fixed and random effect parameters were estimated with good precision with low to moderate variability; The estimate of bioavailability from SC administration is consistent with other biological agents.

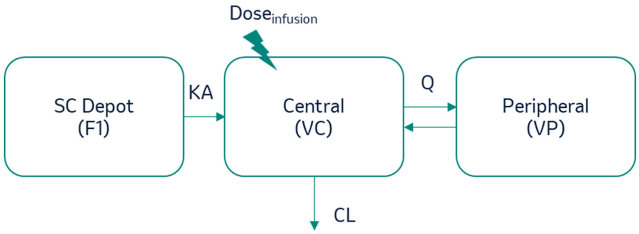

FIGURE 2 Schematic of the population pharmacokinetic model for sotatercept. SC Depot represents the subcutaneous compartment. F1 represents the bioavailability of sotatercept. VC is the central volume of distribution for sotatercept. Q is the intercompartment clearance. VP is the peripheral volume of distribution of sotatercept. CL is the linear clearance of sotatercept from the central compartment.

Covariate analysis

Body weight (BW) was found to be a statistically significant predictor of CL and VC, with typical CL and typical VC predicted to increase with increasing BW. Baseline albumin concentration (ALB) was a statistically significant predictor of CL, with typical CL predicted to decrease with increasing baseline ALB concentration. None of the remaining covariates tested were identified as statistically significant sources of IIV in sotatercept PK. To depict the impact of statistically significant covariate effects on sotatercept PK, individual CL for the phase 2/3 population are plotted vs. BW and baseline ALB (Figure 3).

AUCss are visually depicted in Figure 4 including GMR and associated 90% CIs as a consequence of the influence of statistically significant covariates (body weight and ALB) as well as other relevant intrinsic demographic factors of interest (age, sex, racial classification, and renal function category) and extrinsic factors, such as mono, double, and triple combination with background standard of care (SOC) therapy. Hypothetical SC dosing of 0.7 mg/kg of sotatercept every 21 days was used to obtain model-predicted exposures. The low and high covariate groups for continuous covariates reflect clinically representative distributions of the range of covariate values in this population. For the overall phase 2/3 analysis population as well as the population of patients with PAH from Study MK 7962 003, the GMRs were close to 1 for AUCss with the 90% CI in each case contained within bioequivalence bounds (0.8, 1.25) (Figure 4). Thus, the magnitude of the impact of ALB is not clinically relevant. The impact of BW was taken into account by using weight-based dosing. A median PK profile (along with 2.5th-97.5th

TABLE 2 Parameter estimates with their corresponding percentage of relative standard errors (%RSE) as well as bootstrap means for these parameters with their corresponding 95% CI for the final integrated phase 1/2/3 population pharmacokinetic model.

Parameter, definition and (unit)		Final parameter estimate			Magnitude of variability		
		Population mean	%RSE	Bootstrap mean (95% CI)	Final estimate	%RSE	Bootstrap mean (95% CI)
CL	Central clearance in typical participant (L/day)[a]	0.177[b]	6.22	0.176 (0.150, 0.200)	28.3 %CV	16.9	28.2 %CV (23.1%, 32.8%)
	Exponent of body weight power effect (−)[a]	0.814	11.9	0.809 (0.604, 1.00)			
	Exponent of baseline albumin effect (−)[a]	−0.849	24.2	−0.851 (−1.27, −0.454)			
VC	Central volume (L)[c]	3.60	5.50	3.61 (3.25, 4.11)	24.7 %CV	31.0	24.0 %CV (15.3%, 31.7%)
	Exponent of body weight power effect (−)[c]	1.02	11.8	0.999 (0.732, 1.25)			
Q	Distribution clearance (L/day)	0.487	20.5	0.476 (0.239, 0.677)	NE	NA	NA
VP	Peripheral volume (L)	1.71	15.8	1.66 (0.740, 2.24)	73.3 %CV	28.2	81.9 %CV (52.9%, 188.2%)
KA	First-order absorption rate constant (1/day)	0.273	16.0	0.275 (0.199, 0.371)	60.4 %CV	24.2	60.4 %CV (41.6%, 77.3%)
F1	SC bioavailability (−)	0.659[b]	6.29	0.657 (0.557, 0.745)	11.9 %CV[d]	26.3	11.9 %CV (5.8%, 19.8%)
Residual variability in HV (log units)		0.0570	20.2	0.0574 (0.0376, 0.0838)	0.239 SD[e]	NA	0.240 SD (0.194, 0.289)
Residual variability in PAH (log units)		0.0357	12.6	0.0357 (0.0279, 0.0456)	0.189 SD[e]	NA	0.189 SD (0.167, 0.214)
Minimum value of the objective function = −6707.007; Condition number = 124							

Note: Shrinkage estimates: 17.3% for IIV in CL, 51.9% for IIV in VC, 34.6% for IIV in VP, 48.3% for IIV in KA, and 36.8% for IIV in F1.

Abbreviations: ALB, baseline albumin; CI, confidence interval; %CV, coefficient of variation expressed as a percent; HV, healthy participant; IIV, inter-individual variability; IIV, inter-individual variability; MM, multiple myeloma; NA, not applicable; NE, not estimated; PAH, pulmonary arterial hypertension; PK, pharmacokinetic; %RSE, relative standard error expressed as a percent; RV, residual variability; SC, subcutaneous; SD, standard deviation; WTKGT, time-varying body weight.

[a]The typical value of clearance, $\widetilde{CL}$, for an individual with WTKGT and ALB can be calculated as follows $\widetilde{CL} = 0.177 \times \left(\frac{WTKGT}{70}\right)^{0.814} \times \left(\frac{ALB}{4.5}\right)^{-0.849}$.

[b]The following parameter estimates were found to be highly correlated ($r^{2\,3}$ 0.81): F1: SC bioavailability (−), CL: central clearance in typical participants (L/day).

[c]The typical value of central volume, $\widetilde{VC}$, for an individual with WTKGT can be calculated as follows $\widetilde{VC} = 3.60 \times \left(\frac{WTKGT}{70}\right)^{1.02}$.

[d]The magnitude of IIV expressed as %CV of SC bioavailability (−) was calculated using the following equation: 100 ´ (1–0.659) ´ 0.348. The exponential IIV assumed a log-normal distribution, while a logit IIV model for F1 assumed a normal distribution with a mean of 0 and a variance equal to ω_x^2. The estimates are presented as %CVs approximated by $\%\,CV_x = \sqrt{\exp(\omega_x^2) - 1} \times 100$ for the exponential error model and $\%\,CV_x = (1 - \%\tilde{X}_i) \times \sqrt{\omega_x^2} \times 100$ for the logit error model, with $\tilde{X}_i$ is the typical value of the X parameter in the *i*th participant.

[e]The RV estimate for the log error model is expressed as a standard deviation (SD) (σ) in log-transformed concentration units assumed distributed with a mean of 0 and a variance of σ^2.

inter-percentile range) of the FDA-approved dosing regimen of sotatercept (i.e., 0.3 mg/kg followed by 0.7 mg/kg Q3W) is shown in Figure S1.

Model qualification

Figure 5 shows the pcVPC of sotatercept PK. The median-predicted concentrations tracked the median observed concentrations generally well across studies and dosing intervals. Although concentrations at later times were slightly underpredicted for SAD and MAD studies, the model characterized sotatercept exposures well in patients with PAH (SPECTRA, PULSAR, and STELLAR).

Additionally, a bootstrap analysis was performed, with 99.3% of the bootstrap runs minimizing successfully. From these successful models, the nonparametric 95% bootstrap CIs for each parameter are summarized in Table 2. Each of the parameter estimates from the final PK model fell within the corresponding 95% CI from the bootstrap analysis, supporting a very small or negligible bias in all parameter estimates.

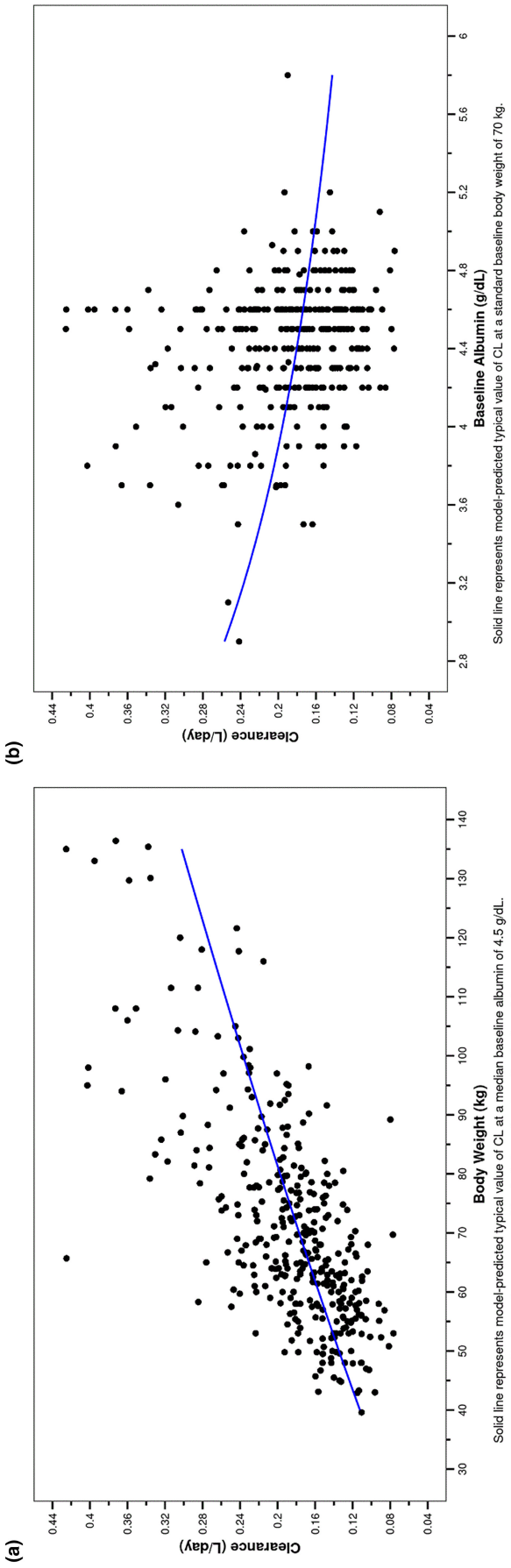

FIGURE 3 Scatterplot of individual Bayesian estimates of clearance from the final integrated phase 1/2/3 population pharmacokinetic model versus (a) body weight and (b) albumin overlaid with a model-predicted line for a typical patient.

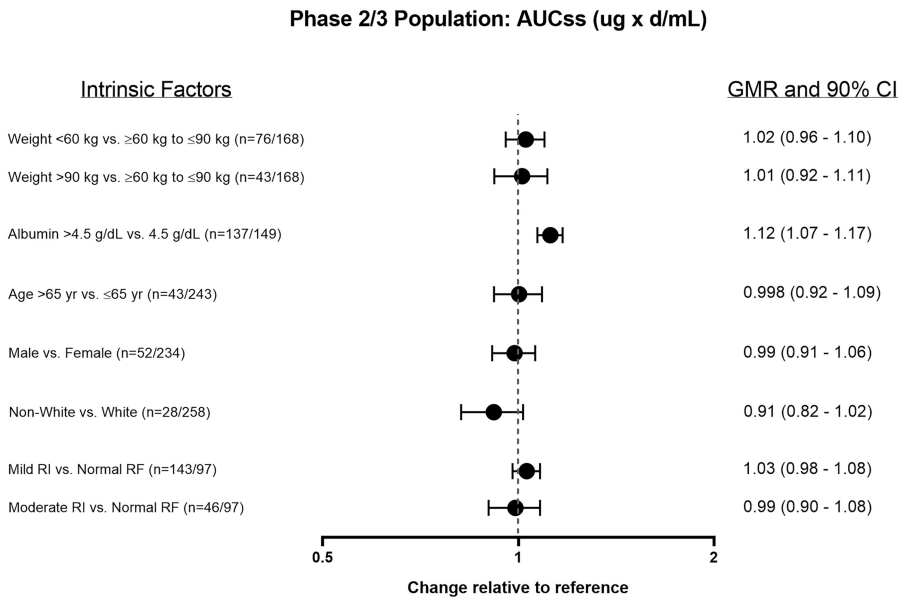

FIGURE 4 Forest plot of geometric mean ratios (90% confidence intervals) of estimated covariate effects on sotatercept exposure following subcutaneous doses of 0.7 mg/kg every 21 days for 24 weeks in the integrated phase 2/3 analysis population. RF is renal function; RI is renal impairment. All statistics are rounded to 2 decimal places, except for the GMR value (0.998) for age.

DISCUSSION

The PK of sotatercept was analyzed using PK data from clinical trials including healthy PMW participants (2 phase 1), and PAH patients (two phase 2 and one phase 3). The phase 1 studies allowed for the characterization of sotatercept PK profile in a wide range of single doses (0.01–3 mg/kg) as well as monthly doses (0.03–1 mg/kg Q4W), administered IV or SC. The phase 2/3 studies allowed for the evaluation of the PK of sotatercept in the target population at 0.3 or 0.7 mg/kg Q3W (phase 2) and at the initial dose of 0.3 mg/kg followed by the target dose of 0.7 mg/kg Q3W thereafter (phase 3).

Sotatercept is a fusion protein, composed of the extracellular domain of the human activin receptor type IIA fused to the Fc domain of human IgG1. The complete mechanism behind its elimination from the body is not fully comprehended, yet it is believed to share similar elimination pathways with monoclonal antibodies (mAbs) owing to its molecular weight and biological structure. At all studied doses, sotatercept PK appeared to be linear as assessed by the dose-normalized exposure data. Therefore, it was assumed that at all tested doses, sotatercept serum concentrations exceeded the minimum target saturation concentration, leading to a saturation of sotatercept major pharmacological target, Activin A, and hence was deemed irrelevant for inclusion in the PopPK model. This hypothesis is supported by the calculated mean $t_{1/2}$ of sotatercept that was predicted to be ~22 days (participants weighing 70 kg with a median normal baseline ALB of 4.5 g/dL), which is comparable to the average $t_{1/2}$ of mAbs.[19]

While most FDA-approved PAH-specific therapeutics are small-molecule drugs that target pulmonary vasoconstrictory and vasodilatory signaling pathways,[20] sotatercept is a first-in-class biologic in the PAH treatment armamentarium with a novel mechanism of action.[15] With a longer half-life (~22 days) than other PAH small-molecule drugs (average $t_{1/2}$ ~ 4–24 h), sotatercept results in less frequent dosing (i.e., Q3W) and consequently may lead to improved patient compliance. This benefit may translate into a more effective therapy with reduced off-target effects.[16]

The current PopPK model was fit to both SC and IV data simultaneously without differentiating between the IIV or RV terms for SC and IV administrations, as the PK variability is not significantly increased by SC administration over IV. Although the model slightly underpredicted the high-sotatercept exposures for healthy participants in MAD (Figure S2), overall, the final PopPK model captured observed data across the five clinical studies reasonably well. The estimated F1 is ~66%, which is consistent with the bioavailability of other biologics such as mAbs,[21] and supports the advancement of SC formulation. Consequently, sotatercept PK profile following SC administration proves advantageous, justifying the decision to exclusively investigate the SC route of administration in both phase 2 and phase 3 trials. This decision was reinforced by the favorable SC PK profile, enhancing patients' convenience.[22]

The covariate analysis identified the time-varying BW as a significant descriptor of both CL and VC; as well as the baseline ALB concentration to be a significant predictor of CL. The population typical values of CL and VC were best

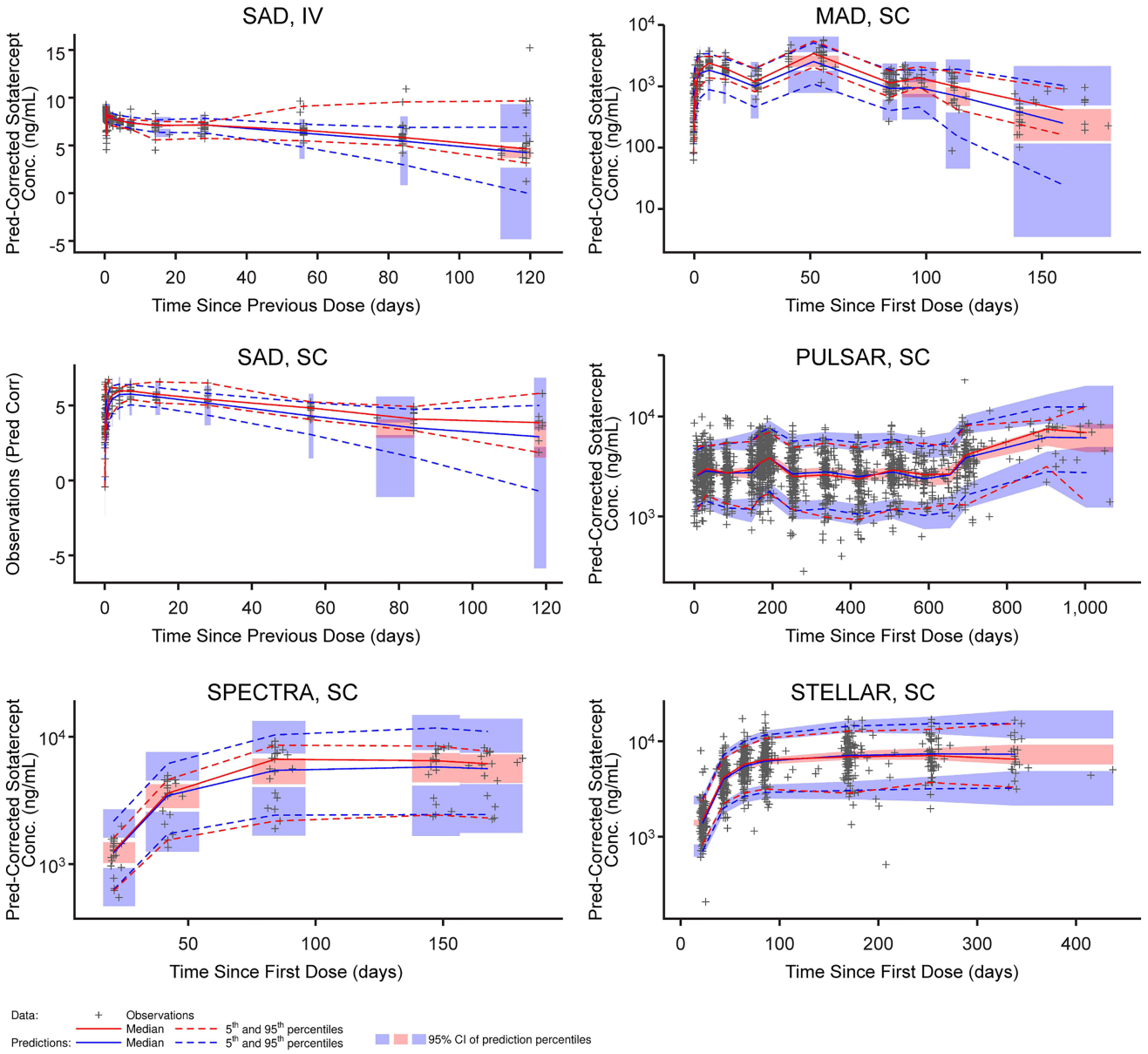

FIGURE 5 Prediction-corrected visual predictive check (pcVPC) of the final integrated phase 1/2/3 population pharmacokinetic model, stratified by study and dosing route. The pcVPC in a linear Y-axis scale is depicted in Figure S2.

described by power functions of BW, where the estimates of the power coefficients (0.81 for CL, 1.0 for VC) were close to the allometry scaling factors.[23,24] They indicate that CL and VC increase with increasing BW. For example, the typical values for CL are 0.135, 0.177, and 0.237 L/day for subjects with BWs of 50, 70, and 100 kg, and the typical values for VC are 2.55, 3.60, and 5.18 L for subjects with BWs of 50, 70, and 100. However, this increase in CL and VC with an increase in BW is generally not clinically important in the context of weight-based dosing.

With an observed 35% variability in sotatercept AUC, a weight-based dosing that is lower than the typical mAb variability, together with a modest clearance variation with weight (0.81 exponent), these results suggest that fixed dosing strategies for sotatercept are likely to have acceptable PK variability. Future investigation of this potential for fixed dosing is warranted. Yet, BW can impact how mAbs are processed in the body through various biologically plausible mechanisms. In instances, such as obesity, decreased lymph flow rates might affect the distribution of mAbs in tissues, influencing both their rate and extent of spread.[25] Moreover, cancer patients with lower BWs, potentially due to cachexia,[26] might experience increased rates of protein endocytosis and breakdown, leading to escalated mAb clearance. Additionally, owing to their relatively large molecular weight (average 150 kDA), mAbs exhibit sluggish tissue uptake and a limited volume of distribution in the central compartment (VC which is

typically around 3–4 L), primarily residing in vascular and interstitial spaces. Consequently, BW is anticipated to correlate with VC as a determinant of mAb distribution. This idea finds support in the argument that larger individuals, possessing a greater volume of plasma and interstitial fluid, tend to have a higher volume of distribution, while smaller individuals exhibit lower linear elimination compared with their larger counterparts.[27]

Covariate analysis also identified that CL of sotatercept decreased with increasing ALB concentrations. However, the overall effect of ALB concentration on CL was relatively modest, with typical CL predicted to decline by 30% from minimum to maximum values of ALB concentrations in the totality of all participants. Albumin, the most abundant protein in the plasma, is primarily synthesized by the liver and is extensively distributed between the intravascular and extravascular spaces. Albumin correlates with BW and may be considered as a derived anthropometric parameter; however, its significance as a covariate may relate to the fact that albumin, like IgG, is protected from intracellular catabolism by FcRn. Unlike IgG, albumin binds to FcRn non-cooperatively at a distinct site.[28] While FcRn has a more pronounced impact on IgG clearance due to IgG's lower concentration relative to albumin, it is estimated that FcRn salvages around 35 times more ALB per unit time compared with IgG (i.e., relative to FcRn salvage of IgG).[29] In clinical trials, baseline serum ALB was identified as a significant covariate explaining IIV in clearance for many mAbs such as Infliximab (in ulcerative colitis[30] and Crohn's disease[31]), ustekinumab,[32] and pertuzumab.[33]

Although sotatercept is a biologic with a large molecular weight, therefore not expected to be filtered by the kidneys, renal function was still tested as a covariate and was not found to be a significant factor with regard to sotatercept exposure. However, it should be noted that severe renal insufficiency (eGFR <30 mL/min/1.73m^2) was an exclusion criterion for the studies included in this analysis, while participants with moderate (eGFR between 30 and 60 mL/min/1.73 m^2) or mild (eGFR between 60 and 90 mL/min/1.73 m^2) impairment were included in the clinical trials, but did not impact sotatercept exposures, which compared with drug exposures from participants with normal renal function (eGFR between 90 and 120 mL/min/1.73 m^2). As a sotatercept is a biologic, renal function is not expected to be relevant in contrast to small-molecule therapies that rely on renal excretion as part of their metabolism.

The impact of other available covariates on sotatercept PK, including age, disease status (i.e. healthy vs. PAH), sex, and racial classification were not found to be statistically significant sources of IIV in sotatercept PK. Of these tested covariates, disease status was included in the model during forward selection but was not retained during the backward elimination step. Furthermore, while immunogenicity of biologics is a clinical concern as they may cause undesirable immunological reactions in some patients, time-varying ADA was not found to be a statistically significant covariate on sotatercept PK. This result is further confirmed by similar PK time profiles, sotatercept C_{trough}, and CL, between participants who were ADA-negative, ADA-positive, and neutralizing antibody (Nab)-negative, ADA-positive, and NAb-positive.[34]

Background PAH therapies in double or triple combination with SOC drugs were not found as statistically significant covariates impacting its PK parameters. This finding suggests that sotatercept is not a victim of drug–drug interaction with background PAH therapies. Neither it is anticipated to affect the PK of concomitantly administered medications such as background PAH therapy.[15] Because sotatercept does not interact with other endogenous targets and is not a proinflammatory cytokine, inflammation-mediated changes in metabolic enzyme function or activity are not expected. This is further supported by preclinical genetic models of PAH showing therapeutic anti-inflammatory activities of three sotatercept analogs that together with its known anti-proliferative effects on vascular cell types, underlie superior clinical activity of sotatercept as add-on therapy to current PAH therapies.[35]

In summary, the final integrated PopPK model captured sotatercept PK from phase 1–3 studies reasonably well. The PK parameters were estimated with reasonable precisions with low to moderate variability. The estimated bioavailability of sotatercept is consistent with other biologics such as mAbs and supports the SC formulation of sotatercept. Time-varying BW was found to impact CL and VC, where typical CL and typical VC were predicted to increase with increasing BW. The typical CL is also predicted to decrease with increasing baseline ALB concentrations. None of the remainders intrinsic and extrinsic covariates tested were identified as statistically significant sources of IIV in sotatercept PK. Model-predicted steady-state sotatercept exposures were not meaningfully changed by the statistically significant covariates, suggesting that the magnitudes of covariate effects do not significantly alter the disposition of sotatercept and are not likely to be clinically relevant.

AUTHOR CONTRIBUTIONS

S.A.O. and F.G. wrote the manuscript with assistance from a medical writer. S.A.O., Z.H., L. W., and F.G. designed the research. S.A.O., D.J., S.B., Z.H., S.H., B.B., B.M., J.O.P., L.W., and F.G. performed the research and analyzed the data.

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CONFLICT OF INTEREST STATEMENT

SAO, ZH, SH, JOP, LW, and FG are employees of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA and may hold stock options in Merck & Co., Inc., Rahway, NJ, USA. DJ is an employee of Cognigen division of Simulations Plus Inc., Buffalo, NY and may hold stock options in Simulation Plus. SB is currently an employee of Regeneron Pharmaceuticals, Inc. and may hold stocks in Regeneron Pharmaceuticals, Inc; SB was previously an employee of Cognigen. BB is currently an employee of Alector LLC, South San Francisco, CA, USA and may hold stock options in Alector LLC. BM is currently an employee of Insmed Incorporated, Bridgewater, NJ, USA and may hold stock options in Insmed. BB and BM were previous employees of Acceleron Pharma, Inc., a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

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REFERENCES

1. Mandras SA, Mehta HS, Vaidya A. Pulmonary hypertension: a brief guide for clinicians. *Mayo Clin Proc*. 2020;95(9):1978-1988.
2. Peacock AJ, Murphy NF, McMurray JJ, Caballero L, Stewart S. An epidemiological study of pulmonary arterial hypertension. *Eur Respir J*. 2007;30(1):104-109.
3. Frost AE, Badesch DB, Barst RJ, et al. The changing picture of patients with pulmonary arterial hypertension in the United States: how REVEAL differs from historic and non-US contemporary registries. *Chest*. 2011;139(1):128-137.
4. Humbert M, Sitbon O, Chaouat A, et al. Pulmonary arterial hypertension in France: results from a national registry. *Am J Respir Crit Care Med*. 2006;173(9):1023-1030.
5. Escribano-Subias P, Blanco I, López-Meseguer M, et al. Survival in pulmonary hypertension in Spain: insights from the Spanish registry. *Eur Respir J*. 2012;40(3):596-603.
6. Ling Y, Johnson MK, Kiely DG, et al. Changing demographics, epidemiology, and survival of incident pulmonary arterial hypertension: results from the pulmonary hypertension registry of the United Kingdom and Ireland. *Am J Respir Crit Care Med*. 2012;186(8):790-796.
7. Jansa P, Jarkovsky J, Al-Hiti H, et al. Epidemiology and long-term survival of pulmonary arterial hypertension in The Czech Republic: a retrospective analysis of a nationwide registry. *BMC Pulm Med*. 2014;14:45.
8. Leber L, Beaudet A, Muller A. Epidemiology of pulmonary arterial hypertension and chronic thromboembolic pulmonary hypertension: identification of the most accurate estimates from a systematic literature review. *Pulmonary Circulation*. 2021;11(1):2045894020977300.
9. Rubin LJ. Primary pulmonary hypertension. *N Engl J Med*. 1997;336(2):111-117.
10. Simonneau G, Galiè N, Rubin LJ, et al. Clinical classification of pulmonary hypertension. *J Am Coll Cardiol*. 2004;43:S5-S12.
11. Lau EMT, Giannoulatou E, Celermajer DS, Humbert M. Epidemiology and treatment of pulmonary arterial hypertension. *Nat Rev Cardiol*. 2017;14(10):603-614.
12. Huertas A, Tu L, Guignabert C. New targets for pulmonary arterial hypertension: going beyond the currently targeted three pathways. *Curr Opin Pulm Med*. 2017;23(5):377-385.
13. Sommer N, Ghofrani HA, Pak O, et al. Current and future treatments of pulmonary arterial hypertension. *Br J Pharmacol*. 2021;178(1):6-30.
14. Machado RD, Aldred MA, James V, et al. Mutations of the TGF-beta type II receptor BMPR2 in pulmonary arterial hypertension. *Hum Mutat*. 2006;27(2):121-132.
15. Humbert M, McLaughlin V, Gibbs JSR, et al. Sotatercept for the treatment of pulmonary arterial hypertension. *N Engl J Med*. 2021;384(13):1204-1215.
16. Hoeper MM, Badesch DB, Ghofrani HA, et al. Phase 3 trial of Sotatercept for treatment of pulmonary arterial hypertension. *N Engl J Med*. 2023;388(16):1478-1490.
17. Beal SL. Ways to fit a PK model with some data below the quantification limit. *J Pharmacokinet Pharmacodyn*. 2001;28(5):481-504.
18. Bergstrand M, Hooker AC, Wallin JE, Karlsson MO. Prediction-corrected visual predictive checks for diagnosing nonlinear mixed-effects models. *AAPS J*. 2011;13(2):143-151.
19. Ryman JT, Meibohm B. Pharmacokinetics of monoclonal antibodies. *CPT Pharmacometrics Syst Pharmacol*. 2017;6(9):576-588.
20. Shah AJ, Beckmann T, Vorla M, Kalra DK. New drugs and therapies in pulmonary arterial hypertension. *Int J Mol Sci*. 2023;24(6):1-39.
21. Sanchez-Felix M, Burke M, Chen HH, Patterson C, Mittal S. Predicting bioavailability of monoclonal antibodies after

subcutaneous administration: open innovation challenge. *Adv Drug Deliv Rev*. 2020;167:66-77.
22. Ait-Oudhia S, Wang YM, Dosne AG, et al. Challenging the norm: a multidisciplinary perspective on intravenous to subcutaneous bridging strategies for biologics. *Clin Pharmacol Ther*. 2024;115(3):412-421.
23. West GB, Brown JH, Enquist BJ. A general model for the origin of allometric scaling laws in biology. *Science*. 1997;276(5309):122-126.
24. West GB, Brown JH, Enquist BJ. The fourth dimension of life: fractal geometry and allometric scaling of organisms. *Science*. 1999;284(5420):1677-1679.
25. Weitman ES, Aschen SZ, Farias-Eisner G, et al. Obesity impairs lymphatic fluid transport and dendritic cell migration to lymph nodes. *PLoS One*. 2013;8(8):e70703.
26. Jeevanandam M, Horowitz GD, Lowry SF, Brennan MF. Cancer cachexia and protein metabolism. *Lancet*. 1984;1(8392):1423-1426.
27. Gill KL, Machavaram KK, Rose RH, Chetty M. Potential sources of inter-subject variability in monoclonal antibody pharmacokinetics. *Clin Pharmacokinet*. 2016;55(7):789-805.
28. Chaudhury C, Brooks CL, Carter DC, Robinson JM, Anderson CL. Albumin binding to FcRn: distinct from the FcRn-IgG interaction. *Biochemistry*. 2006;45(15):4983-4990.
29. Sand KM, Bern M, Nilsen J, Noordzij HT, Sandlie I, Andersen JT. Unraveling the interaction between FcRn and albumin: opportunities for Design of Albumin-Based Therapeutics. *Front Immunol*. 2014;5:682.
30. Fasanmade AA, Adedokun OJ, Ford J, et al. Population pharmacokinetic analysis of infliximab in patients with ulcerative colitis. *Eur J Clin Pharmacol*. 2009;65(12):1211-1228.
31. Fasanmade AA, Adedokun OJ, Blank M, Zhou H, Davis HM. Pharmacokinetic properties of infliximab in children and adults with Crohn's disease: a retrospective analysis of data from 2 phase III clinical trials. *Clin Ther*. 2011;33(7):946-964.
32. Zhu Y, Hu C, Lu M, et al. Population pharmacokinetic modeling of ustekinumab, a human monoclonal antibody targeting IL-12/23p40, in patients with moderate to severe plaque psoriasis. *J Clin Pharmacol*. 2009;49(2):162-175.
33. Wang X, Fan M, Chen X, et al. Intratumor genomic heterogeneity correlates with histological grade of advanced oral squamous cell carcinoma. *Oral Oncol*. 2006;42(7):740-744.
34. Liao K, Mackenzie H, Ait-Oudhia S, et al. The impact of immunogenicity on the pharmacokinetics, efficacy, and safety of sotatercept in a phase 3 study of pulmonary arterial hypertension. *Clin Pharmacol Ther*. 2023;115:478-487.
35. Joshi SR, Liu J, Bloom T, et al. Sotatercept analog suppresses inflammation to reverse experimental pulmonary arterial hypertension. *Sci Rep*. 2022;12(1):7803.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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